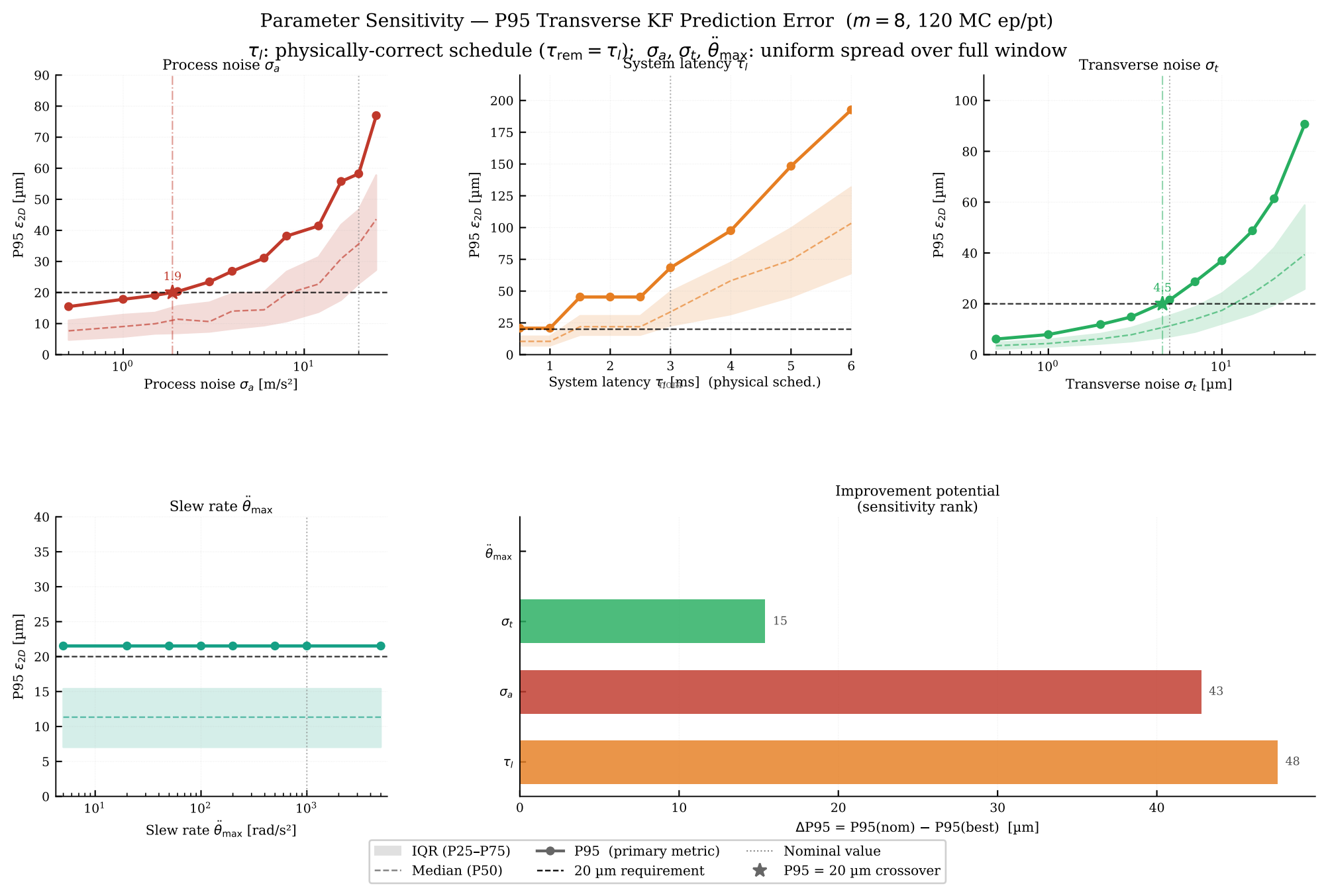
Parameter Sensitivity — P95 Transverse KF Prediction Error (m = 8, 120 MC ep/pt)
τl: physically-correct schedule (τrem = τl); σa, σt, θ̈max: uniform spread over full window
Process noise σa
P95 ε2D [µm]
Process noise σa [m/s²]
19
System latency τl
System latency τl [ms] (physical sched.)
Transverse noise σt
Transverse noise σt [µm]
4.5
Slew rate θ̈max
Slew rate θ̈max [rad/s²]
Improvement potential
(sensitivity rank)
θ̈max
σt
σa
τl
15
43
48
ΔP95 = P95(nom) − P95(best) [µm]
IQR (P25–P75)
Median (P50)
P95 (primary metric)
20 µm requirement
Nominal value
P95 = 20 µm crossover